

Dohyun (Eric) Kim

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EDUCATION

University of Auckland

Expected Graduation Nov 2027

- Degree: *Bachelor of Engineering (Honours) in Computer Systems Engineering*
- Concentrations: Embedded Systems & Hardware Design

EXPERIENCES

Research Assistant

Auckland, New Zealand

University of Auckland

Jul 2026 - Present

- Contribute to the **robot soccer system** at UoA's robotics lab (CARES), including **control and decision making logic**, with future work migrating its classical architecture into a modern software framework.
- Develop and test navigation algorithms for **TurtleBot2**, with sensor integration and performance evaluation.

Robotics Instructor

Auckland, New Zealand

Creative Imaginary Lab (ciLab)

Apr 2026 - Present

- Instructed **50+ students** in robot **hardware assembly and software programming** to complete missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

SELECTED PROJECTS

Flappy Universe

VHDL & FPGA

FPGA Game Implementation - Team Project

Jun 2026

- Implemented a Flappy Bird style game in **VHDL**, using a custom VGA sync generator and PLL based clock division on a **DE0 CV FPGA**, to **render real time sprite based gameplay**.
- Built a **hardware LFSR** random number generator, using **shift register feedback logic**, to procedurally place pipe gaps and drive collision detection and scoring.

Smart Energy Monitor

Embedded C & PCB

Embedded Energy Monitoring Systems - Team Project

Oct 2025

- Designed a dual channel energy monitoring system, using **signal conditioned ADC sampling on an ATmega328P**, to compute RMS voltage, peak current, and average power in real time.
- Built and validated a double layer PCB, using **Altium Designer** and UART based output, to **meet a $\pm 5\%$ full scale accuracy spec** across a 2.5 to 7.5 VA load range.

AWARDED PROJECT

3rd Place in ECSE Design Competition 2025

Team Competition

"Winnie the Bot" - Interactive companion robot

Sep 2025

- Contributed to an AI powered interviewer robot, using **dual ATmega328P Nano** microcontrollers to **drive face tracking, arm movement, and real time voice interaction**.
- **Prototyped and refined the robot's enclosure**, using iterative 3D printing and multiple design revisions, to deliver a compact, durable, and functional physical build.

LEADERSHIPS

Academic Team Executive

Auckland, New Zealand

KEB (Korean Engineering Body) - Founding Member

Jul 2024 - Present

- Developed and delivered tutorial sessions for **20+ junior engineering students** across various courses.
- Participated in the planning and delivery of academic events to **support the student community**.

Official Competition Staff

NZ Robotics Olympiad 2026

IEEE R&A (IEEE Robotics and Automation Society)

Jul 2026

- **Selected as official Competition Staff** based on prior volunteering & robotics instructor experience at ciLab.
- Led competition operations and **resolved technical issues**, ensuring smooth running matches for all teams.

Full-time Student Volunteer

NZ Robotics Olympiad 2025

IEEE (Institute of Electrical and Electronics Engineers)

Jul 2025

- **Volunteered 40+ hours** supporting operations and logistics at NZRO 2025, an IEEE robotics competition.
- Collaborated with organisers to **ensure a smooth event experience** for participants.

SKILLS

Languages & Platforms: C, VHDL, Python, Java, R, MATLAB, ROS, Git, AWS

Hardware & EDA Tools: Altium Designer, LTSpice, ModelSim, Intel Quartus Prime, Proteus, Atmel AVR, AutoCAD